Solar energy - Lecture notes

47202 Introduction to Future Energy

Rafael Nogueira Nakashima (rafnn@dtu.dk)
Technical University of Denmark

Fall 2026

1 The solar energy resource

The Sun is a star in the center of the Solar System formed by the gravitational collapse of a large molecular cloud. In its core, fusion reactions (see Eq. 1) release high levels of energy that heat itself and emits light, similar to an incandescent light bulb.

$$4_1^1H \rightarrow_2^4 He + 2e^+ + 2\nu \qquad Q \approx 4.2 \times 10^{-12}[J] \tag{1}$$

In physics, we can explain light as electromagnetic radiation: a self-propagating wave of electric and magnetic fields that does not require a medium to propagate. Everything at a temperature above the absolute zero (i.e., -273.15 °C) emits "light" as thermal radiation because of the thermal motion of particles (Stefan-Boltzmann law, see Eq. 2). In the ideal case, the body emitting radiation is called blackbody and its emissivity equals one ($\epsilon = 1$).

$$\dot{Q}_{rad}^{out} = \epsilon \sigma A T^4 \tag{2}$$

Symbol	Description	Unit
$\dot{Q}_{rad}^{out}$	Thermal radiation exiting an object	W
σ	Stefan-Boltzmann constant ($5.670374419... \times 10^{-8}$)	W/m^2K^4
ϵ	Emissivity of the object	dimensionless
A	Surface area of the object	m^2
T	Absolute temperature of the object	K

The energy flux received by a surface derived from thermal radiation is called **irradiance** (I, see Eq. 3). From satellite measurements, we know that the average thermal radiation of the Sun per unit of area at the top of Earth's atmosphere is 1361 W/m^2, a value referred to as **solar constant**.

$$I = \frac{\dot{Q}_{rad}^{in}}{A} \tag{3}$$

Symbol	Description	Unit
I	Irradiance	W/m^2
$\dot{Q}_{rad}^{in}$	Thermal radiation incoming to a surface	W
A	Surface area	m^2

Solar resource magnitude

Q. What is the total rate of energy that the Earth receives from the Sun? Assume that the thermal irradiation at the top of Earth's atmosphere is 1361 W/m^2, the Earth's radius is 6.4×10^6 m, and Earth can be approximated as a perfect sphere.

A. We can get an estimate by calculating the thermal radiation incoming to Earth's projected surface in relation to the Sun (i.e., a circle). From Eq. 3:

$$\dot{Q}^{in}_{rad} = AI = \pi R^2 I$$

$$\dot{Q}^{in}_{rad} = \pi \times (6.4 \times 10^6 [m])^2 \times 1361 \left[\frac{W}{m^2}\right]$$

$$\boxed{\dot{Q}^{in}_{rad} \approx 175 \times 10^{15} [W]}$$

Comments. For a frame of reference, the world consumed roughly 186 000 TWh in 2024, which correspond to a bit more than 1 hour of solar irradiance in the globe. Thus, the solar energy represents the main source of energy received by our planet and the centerpiece for reducing our dependency on fossil fuels (i.e., coal, oil, and gas).

At equilibrium, the energy transferred by thermal radiation can have three different outcomes when it reaches another object: (i) it is absorbed by the body ($\dot{E}_{abs}$), (ii) it can be reflected by the body ($\dot{E}_{ref}$), or (iii) it can be transmitted through the body ($\dot{E}_{trans}$). As energy conserves by definition, the sum of absorbed, reflected and transmitted energies should be equal to the incoming thermal radiation (see Eq. 4). The share of energy that is absorbed, reflected, or transmitted are reported as properties of a body[1]: absorptance (α), reflectance (ρ), and transmittance (τ), respectively.

$$\dot{Q}^{in}_{rad} = \dot{E}_{abs} + \dot{E}_{ref} + \dot{E}_{trans} \tag{4}$$

$$\alpha = \frac{\dot{E}_{abs}}{\dot{Q}^{in}_{rad}} \tag{5}$$

$$\rho = \frac{\dot{E}_{ref}}{\dot{Q}^{in}_{rad}} \tag{6}$$

$$\tau = \frac{\dot{E}_{trans}}{\dot{Q}^{in}_{rad}} \tag{7}$$

Symbol	Description	Unit
$\dot{Q}^{in}_{rad}$	Thermal radiation incoming to a surface	W
$\dot{E}_{abs}$	Rate of energy absorbed by the body	W
$\dot{E}_{ref}$	Rate of energy reflected by the body	W
$\dot{E}_{trans}$	Rate of energy transmitted through the body	W
α	Absorptance	dimensionless
ρ	Reflectance	dimensionless
τ	Transmittance	dimensionless

Only a fraction of the direct solar irradiance reaches out the Earth's surface, because part of the energy is either absorbed or reflected by the atmosphere. The fraction of incoming solar radiation that is reflected back into space is called albedo. It is estimated that the yearly global average of albedo is 35%: 20% from clouds, 10% from the cloudless atmosphere, and 5% from the Earth's surface [1]. It is important to note that the greenhouse gases limit the amount of energy emitted by Earth to the

[1]These properties are dependent of the radiation wavelength and direction relative to the object

space. Thus, the greenhouse gases help on regulating the Earth's surface temperature, but in excess lead to a net warming effect.

The ideal case of atmospheric influence would be if the sunlight rays are exactly perpendicular to the Earth's surface, called the **zenith**, what is refereed to as **air mass (AM) factor** 1. However, the Sun is often in within a certain angle relative to the zenith position, which leads to higher losses due to the higher mass of atmosphere that sunlight has to pass to reach the Earth's surface. The AM value can be estimated from the angle difference between the sunlight and the zenith (i.e., the solar zenith angle, z), as described in Eq. 8. The **standard test conditions (STC)** for which solar cell efficiencies are reported is AM 1.5, which correspond to roughly when the sun is 48.2° from the zenith, and irradiance[2] is 1000 W/m^2.

$$AM = \frac{1}{cos(z)} \tag{8}$$

Symbol	Description	Unit
AM	Air mass factor	dimensionless
z	Solar zenith angle	degrees

Irradiance in the surface of the Earth can be measured using an instrument called pyranometer. This measurement is called "global irradiance" if it includes both the direct irradiance emitted by the Sun and the "diffusive sunlight", which consists on the light deflected/scattered by particles in the atmosphere (see Eq. 9). It is important to note that the light scattered forward is often grouped together with the direct sunlight, meaning that "diffusive sunlight" practically refers to "indirect sunlight". Direct and diffuse irradiance can also be measured separately. Direct irradiance can be measured by a pyroheliometer attached to a tracking system, while diffuse irradiance can be measured with a pyranometer by blocking the radiation coming from the solar disk.

$$I_{global} = I_{direct} + I_{diffuse} \tag{9}$$

The irradiance can also be measured in a horizontal plane (i.e., normal to Earth's surface) or in a tilted plane. The direct irradiance is maximum when the plane is completely perpendicular to sunlight, which is called direct normal irradiance (I_{direct}^{normal}). The direct normal irradiance can be used to estimate the irradiance for a plane with an arbitrary angle difference between the plane normal vector and the sunlight direction (see Eq. 10). In the case of a horizontal plane, the angle difference is the **solar zenith angle** (z).

$$I_{direct}^{\theta} = I_{direct}^{normal} cos(\theta) \tag{10}$$

Symbol	Description	Unit
I_{global}	Global irradiance	W/m^2
I_{direct}	Direct irradiance	W/m^2
$I_{diffuse}$	Diffuse irradiance	W/m^2
I_{direct}^{θ}	Direct irradiance at a surface with θ inclination	W/m^2
I_{direct}^{normal}	Direct irradiance at a surface perpendicular to sunlight	W/m^2
θ	Inclination between the surface and sunlight	degree

Horizontal plane vs. normal plane irradiance

Q. Estimate the irradiance of a horizontal plane under conditions analogous to STC (AM1.5, $I_{global}^{normal} = 1000\ W/m^2$)? Assume that direct normal irradiance for a horizontal plane is 900 W/m^2.

[2]It includes both the direct normal and diffuse contribution, more on this later

A. From the definition of global irradiance, Eq. 9:

$$I_{global} = I_{direct} + I_{diffuse} \implies I_{diffuse} = 1000 \left[\frac{W}{m^2}\right] - 900 \left[\frac{W}{m^2}\right]$$

$$I_{diffuse} = 100 \left[\frac{W}{m^2}\right]$$

Now, we can calculate the direct irradiance on a horizontal plane from Eq. 10 and the definition of AM (Eq. 8):

$$I_{direct}^{horizontal} = I_{direct}^{normal} cos(z) = \frac{I_{direct}^{normal}}{AM} = \frac{900}{1.5} \left[\frac{W}{m^2}\right]$$

$$I_{direct}^{horizontal} = 600 \left[\frac{W}{m^2}\right]$$

Thus, the global horizontal irradiance is:

$$I_{global} = I_{direct} + I_{diffuse} = 600 + 100 \left[\frac{W}{m^2}\right]$$

$$\boxed{I_{global} = 700 \left[\frac{W}{m^2}\right]}$$

Comments. As it can be observed, the difference between a horizontal plane and a normal plane can be significant depending on the solar zenith angle (z), which varies depending on the hour and day for a specific location. Thus, one can harvest more solar energy by using a solar tracking system or by finding the optimal tilt angle for a specific location over a reference year.

The geographical and temporal differences in global horizontal irradiance under clear sky conditions are derived from the obliquity of the incoming solar radiation. As previously mentioned, the angle between sunlight and the zenith affects both the atmospheric attenuation, as well as the effective area covered by the sunlight (i.e., projection effect, Eq. 10). These differences are determined by the latitude, the day-night cycle (i.e., Earth's rotation), and the seasonal cycles (i.e., the Earth's orbit).

In practice, we do not always have a "clear sky" because of the weather (i.e., clouds), which affects the final value of average irradiance for each region. For instance, it is noticeable the deserts can have higher irradiance compared to forests although positioned at higher latitude, because forests have much higher precipitation levels than deserts. Thus, we often rely on local measurements or complex models (e.g., Global Solar Atlas) to estimate the solar energy potential of a certain region.

The solar energy potential of a region is often reported as the total energy transferred by sunlight during a certain period (often an average day or year). This measure is called **irradiation** ($\bar{I}$) and it can be obtained by recording (or simulating) the irradiance over a whole period and integrating it over time (Eq. 11).

$$\bar{I} = \int_0^t I dt \approx \sum_i I_i \Delta t_i \tag{11}$$

Symbol	Description	Unit
I	Irradiance	W/m^2
$\bar{I}$	Irradiation	W.h/m^2
t	Time	h
I_i	Average irradiance over interval i	W/m^2
Δt_i	Duration of interval i	h

Mean irradiation differences among countries

Q. Estimate the required horizontal area for collecting 30 kWh per day of solar energy (i.e., an average household) for different countries, based on median values provided by the Global Solar Atlas, reproduced in the table below:

Country	Global horizontal daily irradiation (median)	Latitudes
Denmark	2.76 $\frac{kWh}{m^2}$	54°N - 58°N
Spain	4.64 $\frac{kWh}{m^2}$	36°N - 43°N
Yemen	6.57 $\frac{kWh}{m^2}$	12°N - 20°N
Venezuela	5.40 $\frac{kWh}{m^2}$	0°N - 12°N

A. From the definition of irradiance, Eq. 3, we can determine the area as:

$$A = \frac{\dot{Q}^{in}_{rad}}{I} \tag{12}$$

If we multiply numerator and denominator by time, we can get an analogous equation with irradiation:

$$A = \frac{\dot{Q}^{in}_{rad}}{I} \times \frac{\Delta t}{\Delta t} = \frac{30\,[kWh]}{\bar{I}} \tag{13}$$

Thus, substituting the values for global horizontal irradiance from the table into the previous equation, we have:

Country	Area (median)
Denmark	10.9 m^2
Spain	6.46 m^2
Yemen	4.57 m^2
Venezuela	5.56 m^2

Comments. The solar irradiation can be quite different for each country depending on its latitude and weather. For instance, Yemen requires less than half of area compared to Denmark to collect the same amount of solar energy. This affects the final cost of solar energy, as more area requires more land and panels/collectors. It is also important to note that the relationship with latitude is not linear, but rather peaks close to 22° of latitude (where most deserts also are located).

2 Solar thermal energy

Heat is the simplest form of end-use for solar energy. It consists on using thermal radiation from the Sun to fulfill a task that otherwise would require us to use another source of energy more expensive and/or less renewable (i.e., electricity or fuels). For instance, sunlight can be used to heat houses and reduce the consumption of utilities during winter. By correctly engineering windows that both transmit sunlight and isolate buildings, we can maximize the net energy absorbed from sunlight into our homes. Similarly, one can engineer a reflective paint to cover roofs and reduce the energy absorbed by homes in warm climates, saving electricity from air conditioning.

From thermodynamics, we know that heat can be converted into work by using a thermal cycle. This has the main advantage of using the same technology developed for traditional power plants and engines, as they also are thermal cycles. However, this also bounds the maximum conversion efficiency to the second law of thermodynamics, often refereed as Carnot's efficiency/theorem (see Eq. 14).

$$\eta_{Carnot} = \frac{\dot{W}}{\dot{Q}_{hot}} = 1 - \frac{T_{cold}}{T_{hot}} \tag{14}$$

Symbol	Description	Unit
η_{Carnot}	Thermal engine efficiency limit	dimensionless
$\dot{W}$	Work production rate	W
$\dot{Q}_{hot}$	Heat transfer rate from hot source	W
T_{cold}	Temperature of the cold reservoir	K
T_{hot}	Temperature of the hot reservoir	K

Minimum collector area from solar thermal energy

Q. Suppose that a power plant based on solar thermal energy needs to be designed to produce 1 MW. If the highest temperature in the sunlight collector is 673 K, the environmental temperature is 298 K and direct solar irradiance is 800 W/m^2, what is the minimal collector area size requirement? Assume that all irradiance is absorbed by the collector and there are no heat losses.

A. Here, from the solar irradiance definition (Eq. 3) we can calculate the heat provided by the Sun (hot source) as a function of the collector area:

$$I = \frac{\dot{Q}^{in}_{rad}}{A} \implies \dot{Q}^{in}_{rad} = IA$$

From the Carnot efficiency equation (Eq. 14), we can calculate the required hot heat source as a function of the design power generation and thermal cycle temperatures. In the best case scenario, the hot and cold temperature will equal the collector and environmental temperature, respectively.

$$\eta_{II} = \frac{\dot{W}}{\dot{Q}_{hot}} \implies \dot{Q}_{hot} = \frac{\dot{W}}{\eta_{II}}$$

In which the efficiency can be calculated from the given temperature limits:

$$\eta_{II} = 1 - \frac{T_{cold}}{T_{hot}} = 1 - \frac{298\ [K]}{673\ [K]} = 0.557$$

By equaling the hot heat source to the thermal radiation, we have:

$$A = \frac{\dot{W}}{I \cdot \eta_{II}} = \frac{10^6\ [W]}{800\ [W/m^2] \cdot 0.557}$$

$$\boxed{A \approx 2244\ [m^2]}$$

Comments. In the best case scenario, one requires more than 11 tennis courts (singles = 195.65 m^2) to produce 1 MW of electricity. In reality, there are many inefficiencies which makes this number much larger: (i) losses heat absorbed (ii) losses in heat loss (iii) losses in the thermal cycle (iv) weather and day-night cycles. For instance, the famous solar thermal power plant Crescent Dunes (US) requires 10 909 m^2/MW, roughly 5 times more than in the exercise.

One may propose to increase the efficiency of solar thermal power plants by increasing the collector temperature, as indicated by the Carnot's efficiency equation (Eq. 14). However, all matter emits thermal radiation proportional to its temperature (Stefan-Boltzmann law, Eq. 2), so increasing the collector temperature also increases its heat losses due to thermal radiation. For instance, assuming that emissivity and absorptance for the collector are equal and independent on wavelength ($\epsilon = \alpha$), the net thermal radiation absorbed by a solar collector ($\dot{Q}_{net}$) can be approximated as Eq. 15. Note that $\dot{Q}_{net}$ always decreases with an increase in the collector temperature. In sum, increasing the collector temperature may increase the efficiency of heat conversion (Carnot's efficiency consequence), but reduce the net energy collected by thermal radiation (Stefan-Boltzmann law consequence).

$$\dot{Q}_{net} = \dot{Q}_{direct} + \dot{Q}_{env.} - \dot{Q}_{col.} = \alpha A_{col.} \left(I_{direct} + \sigma(T_{env.}^4 - T_{col.}^4) \right) \tag{15}$$

Symbol	Description	Unit
$\dot{Q}_{net}$	Net thermal radiation absorbed	W
$\dot{Q}_{direct}$	Direct thermal radiation absorbed from Sun	W
$\dot{Q}_{env.}$	Thermal radiation absorbed from environment	W
$\dot{Q}_{col.}$	Thermal radiation emitted by collector	W
$A_{col.}$	Area of the solar collector	m^2
T_{env}	Environment temperature	K
$T_{col.}$	Solar thermal collector temperature	K
I_{direct}	Direct irradiance	W/m^2
α	Absorptance	dimensionless
σ	Stefan-Boltzmann constant ($5.670374419... \times 10^{-8}$)	W/m^2K^4

There are two main workarounds to minimize the heat losses in solar thermal power: (i) use materials with emissivity/absorptances sensitive to wavelengths and (ii) use reflective surfaces to concentrate solar radiation and minimize the collector area. The first solution leverages on the observation that most energy in solar radiation is transferred by short wavelengths because of its high temperature (more on this in the next section). Thus, we can engineer selective coatings that have very high ϵ & α for short wavelengths, but very low ϵ & α for long wavelengths in which heat losses occur. This increases the incoming energy transfer, while reducing the net heat losses (see Eq. 16).

$$\dot{Q}_{net} = A_{col.} \left(\alpha_{sw} I + \epsilon_{lw} \sigma(T_{env.}^4 - T_{col.}^4) \right) \tag{16}$$

Symbol	Description	Unit
α_{sw}	Absorptance for short wavelength radiation	dimensionless
ϵ_{lw}	Emissivity for long wavelength radiation	dimensionless

Finally, the second solution is often called as concentrated solar power. It consists on increasing the incoming solar radiation by providing a large reflective area[3] ($A_{ref.}$) which concentrates heat in a smaller collector area ($A_{col.}$). Similarly with the first solution, this increases the amount of incoming heat in relationship with the losses, which improves the net heat absorption (see Eq. 17).

$$\dot{Q}_{net} = \alpha_{sw} A_{ref.} I + \epsilon_{lw} A_{col.} \sigma(T_{env.}^4 - T_{col.}^4) \tag{17}$$

Symbol	Description	Unit
$A_{ref.}$	Reflective area of the solar concentrator	m^2

3 Photovoltaics

Empirical evidence have demonstrated that light can cause objects to emit electrons, a process called photoelectric effect. In parallel, physicists noted that the energy transmitted by electromagnetic radiation follows very specific distributions of frequencies/wavelengths, which could not be predicted accurately from classic physics (see ultraviolet catastrophe) . These seemly unrelated problems lead to the postulation and discovery of photons, an elementary particle representing a finite energy quantity (quantum) of the electromagnetic field. In other words, electromagnetic waves (i.e., light) transfer energy as photons with discrete amounts of energy (i.e., quantum) proportional to its frequency (see Eq. 18):

$$E = h \cdot f \tag{18}$$

[3]The area depends on the basis in which the irradiance is given

Symbol	Description	Unit
E	Energy	J
h	Planck constant ($6.62607015 \times 10^{-34}$)	J.s
f	Frequency	1/s

As electromagnetic waves propagate at light speed (c = 299 792 458 m/s), we can relate its frequency to its wavelength as described in Eq. 19.

$$\lambda = \frac{c}{f} \tag{19}$$

Symbol	Description	Unit
λ	Wavelength	m
c	Light speed in vacuum (299 792 458)	m/s

The connection between light and electrons through photons meant that one could use sunlight to directly produce electricity, instead of relying on thermal engines. This idea lead to the development of photovoltaic (PV) cells. In a PV cell, light is absorbed by a material called semiconductor and creates a pair of electron and a "hole" (i.e., a vacancy created by the electron displacement, which can be understood as a positive free charge). These charges are separated to opposite poles of the PV cell due to an internal electric field, which exist due to the junction of two different semiconductor materials (i.e., p-n junction). Finally, by connecting the opposite poles of a PV cell, electric current can be produced and used to provide electrical energy. In sum, a PV cell working principle can be divided into three processes: (i) charge generation, (ii) charge separation, and (iii) charge collection.

One important aspect of the charge generation process is that only photons above a certain energy level, the band gap, are able to generate a electron-hole pair. In addition, the energy provided above the band gap level is loss by the electron as heat. Thus, a good PV cell material should not have a band gap too high or too low. If too high, most photons will not be absorbed, whereas if too low, most of photon's energy will be converted into heat. The theoretical optimal band gap maximizing power generation from terrestrial sunlight (single junction material) is estimated to be 1.34 eV (approx. 33 % efficiency) [2]. In practice, most PV cells use silicon which has a band gap of 1.14 eV at room temperature.

Minimum wavelength for silicon

Q. If the bandgap for silicon is 1.12 eV, what is the maximum wavelength in μm a photon can have to be able to create an electron-hole pair?

A. By combining equations 18 and 19 we can write down the photon energy as a function of the wavelength:

$$E_{photon} = h\frac{c}{\lambda}$$

The energy of the photon needs to be higher than the bandgap energy to create an electron-hole pair:

$$E_{photon} \geq E_{bandgap} \implies h\frac{c}{\lambda} \geq E_{bandgap}$$

$$\lambda \leq \frac{h \cdot c}{E_{bandgap}}$$

The bandgap in SI units is:

$$E_{bandgap} = 1.14\ [eV] \approx 1.14\ [eV] \times 1.6 \cdot 10^{-19}\ [J/eV]$$

$$E_{bandgap} \approx 1.82 \cdot 10^{-19}\ [J]$$

Now substituting constants and the bandgap value

$$\lambda \leq \frac{6.63 \cdot 10^{-34} \times 3 \cdot 10^{8}}{1.82 \cdot 10^{-19}}\ [m]$$

$$\lambda \leq 1.09 \cdot 10^6 \ [m]$$

$$\boxed{\lambda \leq 1.09 \ [\mu m]}$$

The band gap exist as a consequence of the discrete nature of energy: electrons in a solid can only exist on certain energy levels called electronic bands. On conductive materials, the band gap is almost inexistent and electrons can easily access higher levels of energy and move across a solid. On the other hand, insulator materials have completely full electronic bands with a large band gap to the next electronic band. Therefore, electrons can not easily access higher energy levels, making them practically immovable. Semiconductors are materials in between these extremes, in which the band gaps are potentially accessible if sufficient energy is provided to the electron (e.g., by a photon or temperature).

Semiconductor material characteristics can be enhanced by adding specific impurities, a process called doping. For instance, phosphorous atoms can "fit" on a silicon crystalline structure but they have one extra electron, which adds a new degree of freedom to the movement of charges. On the other hand, an electron vacancy (i.e., a hole) can be added to the silicon crystalline structure by adding boron. Semiconductor materials can be enhanced by doping atoms of either higher or lower atomic numbers, the first is called n-type while the second is called p-type. When p-type and n-type semiconductor materials are put together, the free charges (i.e., electrons and holes) naturally tend to cancel each other leaving an intrinsic electric field from the fixed charges, which separates new electron-hole pairs created by sunlight in a specific direction for charge collection. A complete explanation of how and why this process occur would require us to explain quantum physics, which is beyond the scope of this introduction to solar energy.

Electric current can be generated by connecting a load between the terminals of a photovoltaic (PV) cell, thereby closing the electrical circuit. The magnitude of the current depends primarily on the incident irradiance and the operating voltage of the cell. The relationship between current and voltage is known as the current-voltage (I-V) **characteristic curve**. In an ideal solar cell, the current remains nearly constant under reverse-bias conditions, where the p-type region is at a lower electrical potential than the n-type region. Reverse bias strengthens the electric field within the p-n junction, enhancing charge separation and limiting changes in current. Under forward-bias conditions, the internal electric field of the p-n junction is reduced. As a result, charge carriers can more easily cross the junction, producing a diode current that opposes the photocurrent generated by the incident light. As the forward voltage increases, the net output current decreases, eventually reaching zero at the open-circuit voltage. In the ideal case, we can describe the characteristic curve as an exponential relationship, as denoted in Eq. 20[4].

$$j = j_{SC} - j_0 \left(\exp\left(-\frac{F}{R}\frac{V}{T} \right) - 1 \right) \tag{20}$$

Symbol	Description	Unit
V	Voltage	V
j	Current density	A/cm^2
j_{sc}	Short circuit current density	A/cm^2
j_0	Dark saturation current	A/cm^2
F	Faraday constant (96 485.332 12...)	A.s/mol
R	Universal gas constant (8.314 462 618...)	J/K.mol
T	Temperature	K

Similar to other electric devices, the specific electric power (W/m^2) of an photovoltaic cell can be calculated by multiplying the voltage and current:

$$p = V \cdot j \tag{21}$$

[4]Here a positive power ($V \cdot j$) means energy "entering" the cell (i.e., power consumption), while a negative power means energy "exiting" the cell (i.e., power generation)

Symbol	Description	Unit
p	Specific electric power	W/m^2
V	Cell voltage	V
j	Current density	A/m^2

The maximum power generation is lower than the simple multiplication of maximum voltage (i.e., open circuit voltage, V_{OC}) and maximum current (i.e., short-circuit current j_{SC}). The fill factor is a performance indicator for solar cells that indicate how close a cell is of achieving the power limit and can be described as Eq. 22. A commercial silicon PV cell has an average fill factor of 80 % [2].

$$FF_{max} = \frac{p_{max}}{V_{oc} \cdot j_{sc}} \tag{22}$$

Symbol	Description	Unit
FF	Fill factor	dimensionless
p_{max}	Maximum specific power generation	W/m^2
V_{oc}	Open circuit voltage	V

It is important to highlight that fill factor and efficiency are different concepts. The efficiency measures the ratio of power generation by the total energy provided to the solar cell, as denoted in Eq. 23. In general, new commercial solar cells achieve 20-25% efficiency in converting irradiance into electricity without counting losses in electric components [2]. Moreover, differently from solar thermal power, higher PV cell temperatures reduce the efficiency of power generation. For instance, one can observe that higher temperatures lead to rapid decrease of current in Eq. (20), lowering the maximum power generation. It is estimated that every degree increase in temperature lead to a reduction of 0.4%-0.5% in silicon PVs efficiency [1].

$$\eta_{pv} = \frac{p}{I_{global}} \tag{23}$$

Symbol	Description	Unit
η_{pv}	Solar cell efficiency	dimensionless
I_{global}	Global irradiance reacting the solar cell surface	W/m^2

Efficiency vs. fill factor

Q. The table below shows two hypothetical PV pannels. Which one is more efficient?

Data	A type	B type
Area	1 m^2	1 m^2
Fill factor	0.8	0.7
Power	175 W	250 W
Irradiance	800 W/m^2	1000 W/m^2

A. Here we should be carefull to not conclude which one is more efficient without calculating the efficiency for both cases. From the definition of efficiency we have:

$$\eta_{pv} = \frac{p}{I_{global}} = \frac{P_{pv}/A_{pv}}{I_{global}}$$

For A:

$$\eta_{pv,A} = \frac{175/1}{800} \approx 0.22$$

For B:

$$\eta_{pv,B} = \frac{250/1}{1000} = 0.25$$

Therefore:

$$\boxed{\eta_B > \eta_A}$$

Comment. Although A has a higher FF, the PV pannel B is more efficient.

References

[1] Bent Sørensen. Chapter 2 - origin of renewable energy flows. In *Renewable Energy (Fourth Edition)*, pages 35–204. Academic Press, Boston, fourth edition edition, 2011.

[2] Wikipedia contributors. Solar-cell efficiency — Wikipedia, the free encyclopedia, 2026. [Online; accessed 21-June-2026].